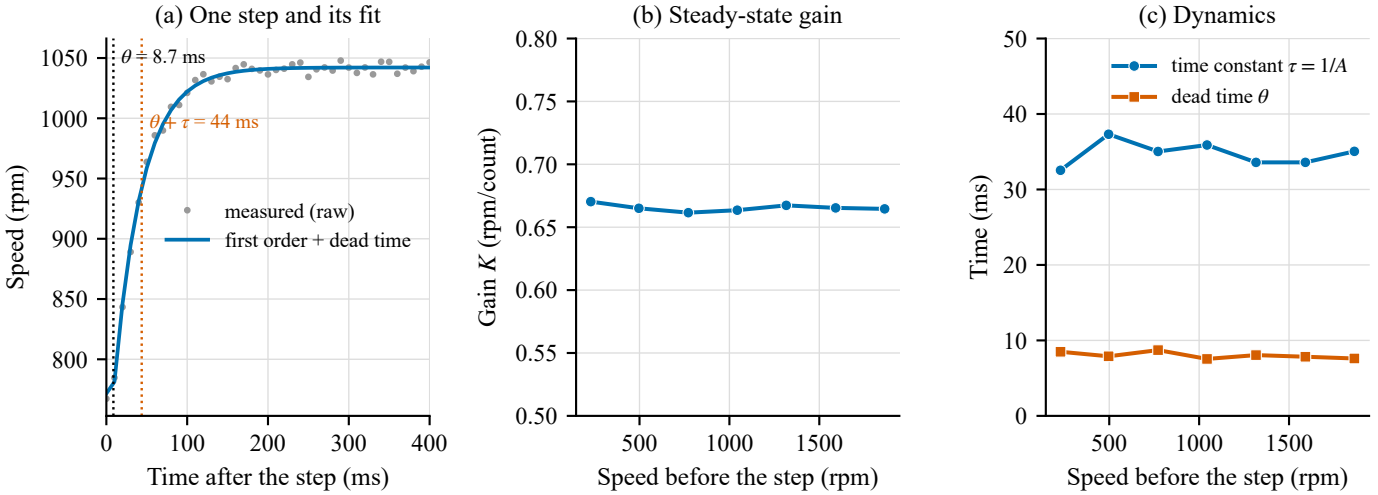
(a) One step and its fit
θ ≈ 8.7 ms
θ + τ = 44 ms
measured (raw)
first order + dead time
Speed (rpm)
Time after the step (ms)
(b) Steady-state gain
Gain K (rpm/count)
Speed before the step (rpm)
(c) Dynamics
time constant τ = 1/A
dead time θ
Time (ms)
Speed before the step (rpm)